RS-SOH: Benchmarking Remote Sensing Small Object Hallucination in Multimodal Large Language Models

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ABSTRACT

Reliable small-object understanding in high-resolution remote sensing imagery is essential for many real-world applications, including traffic monitoring, disaster response, and security inspection. However, with the rapid development of remote-sensing multimodal large language models (RS-MLLMs), it remains unclear whether these models can ground their answers in small-object evidence when visual cues are weak. Existing studues mainly focus on broad scene understanding and coarse object recognition, leaving small-object reliability insufficiently examined. To address this gap, we introduce Remote Sensing Small Object Hallucination (RS-SOH), a benchmark for evaluating small-object hallucination in RS-MLLMs. RS-SOH is constructed from large-scale remote sensing imagery, and contains 25,000 questions generated from 1,000 image patches and 28 small object categories, with objects defined by a bounding-box area smaller than 32×32 pixels. The benchmark covers three complementary reasoning tasks: existence reasoning for object-presence judgment, counting reasoning for numerical verification and instance enumeration, and positional reasoning for spatial grounding at both region and object-relation levels. Experiments on representative RS-MLLMs using RS-SOH show that models exhibit different reliability limitations across reasoning tasks. When visual evidence is weak, RS-MLLMs often make unstable object-presence judgments. Their counting performance is limited by weak instance separation and unreliable numerical verification. Positional reasoning remains difficult because models struggle to ground objects to image regions or object-to-object spatial relations. Scale-aware analysis shows that counting accuracy is more directly related to object size, whereas existence and positional reasoning are less consistently controlled by scale. These findings identify small-object hallucination as a central reliability challenge in remote-sensing multimodal reasoning and suggest that RS-SOH can serve as a practical diagnostic tool for developing more reliable remote-sensing foundation models.

1. Introduction

Remote sensing multimodal large language models (RS-MLLMs) have recently emerged as a promising framework for remote sensing image understanding. By integrating visual perception with language reasoning and instruction-following capabilities, RS-MLLMs allow users to query remote-sensing images in natural language and obtain responses that combine visual recognition with semantic interpretation (Bazi et al., 2024; Kuckreja et al., 2024). Their application potential has been examined in several task-oriented remote-sensing scenarios. For disaster and damage assessment, models need to interpret post-event imagery and connect visible scene changes with affected areas after extreme events (Wang et al., 2025b). Urban and land-use analysis places greater emphasis on object-relation reasoning, since planning-related interpretation often depends on linking buildings, roads, and land-use patterns to semantic descriptions of the built environment (Wang et al., 2024b,a). Agricultural and environmental applications provide another test case: models are expected to recognize land-cover patterns, crop-related cues, and other monitoring-relevant information from remote-sensing imagery (Lobry et al., 2020; Li et al., 2025). These task-oriented settings require RS-MLLMs to support decision-relevant visual reasoning rather than only general image description. This requirement makes reliability a central issue for practical remote-sensing applications. Hallucination is a particularly critical reliability risk. Here, hallucination refers to responses that are not supported by image evidence (Rohrbach et al., 2018). Such errors may lead to false alarms, missed targets, and incorrect judgments in remote sensing applications, thereby affecting downstream analysis and decision-making.

This risk is especially pronounced in small-object understanding. In remote sensing images, many practically important targets, such as vehicles and other man-made objects, are visually subtle, densely distributed, and easily

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confused with surrounding clutter. Compared with larger and more visually salient targets, small objects provide weaker and more ambiguous evidence for recognition (Yu et al., 2020). When such evidence is insufficient, RS-MLLMs may rely more on prior expectations learned from language than on the image itself, which increases the likelihood of unsupported responses (Agrawal et al., 2018). For RS-MLLMs, hallucination in small-object scenarios represents a specific reliability concern that is still insufficiently examined in current benchmark studies.

Current hallucination evaluations do not fully address this setting. General-domain benchmarks are mainly designed for natural images and therefore do not capture the overhead perspective, dense layouts, and mixed object-background structures common in remote sensing imagery. Meanwhile, most RS-MLLM studies still emphasize visual-language capability rather than reliability under weak visual evidence. Although recent work has begun to examine hallucination in RS-MLLMs (Liu et al., 2026; Zhou et al., 2026), the specific failure modes caused by small objects remain insufficiently understood.

To fill this gap, we introduce Remote Sensing Small Object Hallucination (RS-SOH), a benchmark for evaluating small-object hallucination in RS-MLLMs. Rather than reusing scene-level captions, RS-SOH converts annotated small-object records into diagnostic questions that test object presence, quantity judgment, and spatial grounding. Using RS-SOH, we evaluate representative open-source RS-MLLMs and analyze how object scale affects their reliability across different reasoning requirements.

In summary, contributions of our research can be outlined as follows:

- We introduce RS-SOH, a benchmark for evaluating small-object hallucination in RS-MLLMs. RS-SOH contains 25,000 question–answer pairs from 1,000 remote-sensing image patches and covers 28 retained object categories, with all annotated targets smaller than 32×32 pixels. The benchmark covers existence reasoning, counting reasoning, and positional reasoning, explicitly targeting the challenges posed by small-object scale and aerial perspectives.

- We conduct a systematic evaluation of representative RS-MLLMs using this benchmark, revealing prevalent hallucination behaviors and diverse failure modes in small-object understanding, with substantially larger performance disparities across models in counting reasoning and positional reasoning than in existence reasoning.

- We present a scale-aware analysis of hallucination behaviors, showing that object scale affects different reasoning capabilities in a task-dependent manner: counting performance exhibits a clear and significant dependence on object scale, while existence and positional reasoning show weaker or model-dependent sensitivity to object scale.

2. Related Work

2.1. Hallucination in multimodal large language models

Hallucination in multimodal large language models generally refers to outputs that are not grounded in visual evidence, such as mentioning nonexistent objects, assigning incorrect attributes, or making spatial claims inconsistent with the image. Early studies mainly examined hallucination in image captioning and introduced object-centric metrics such as CHAIR (Rohrbach et al., 2018). Later work such as ALOHa (Petryk et al., 2024) further extended this line toward open-vocabulary evaluation. With the rise of MLLMs, hallucination assessment moved beyond captioning to more diagnostic settings. Representative benchmarks include POPE (Li et al., 2023) and NOPE (Lovenia et al., 2024), which probe object-existence hallucination through yes/no or visual question answering-style questions, as well as more recent benchmarks such as ODE (Tu et al., 2025), HallusionBench (Guan et al., 2024), AMBER (Wang et al., 2023), and Hal-Eval (Jiang et al., 2024), which extend evaluation to more diverse error types and reasoning settings. Together, these studies have substantially improved our understanding of hallucination in general-domain MLLMs.

Existing hallucination benchmarks are still mainly designed for natural images. They usually focus on object presence, attribute correctness, or general reasoning errors, rather than the challenges that arise in remote sensing imagery. In particular, they rarely examine the effects of very small objects, dense layouts, or spatial ambiguity under overhead observation. They also do not clearly separate hallucination into reasoning types that are especially important in remote sensing, such as existence, counting, and positional reasoning. As a result, these benchmarks are not sufficient for fully understanding how RS-MLLMs fail on small objects in remote sensing images.

2.2. Remote-sensing multimodal large language models

Recent years have seen rapid development in RS-MLLMs. Early efforts such as RSGPT (Hu et al., 2025b), RS-LLaVA (Bazi et al., 2024), and EarthGPT (Zhang et al., 2024) mainly adapted general vision-language models to remote sensing and focused on captioning, visual question answering, and general scene understanding. Later, the field moved to broader and more fine-grained settings. GeoChat (Kuckreja et al., 2024) introduced grounded dialogue for remote sensing, while SkyEyeGPT (Zhan et al., 2025) and SkySenseGPT (Luo et al., 2024) pushed instruction tuning and unified remote sensing understanding to a larger scale. After that, EarthDial (Soni et al., 2025), TEOChat (Irvin et al., 2025), and EarthMind (Shu et al., 2025) further extended RS-MLLMs to multi-sensor, temporal, and multi-granular Earth observation understanding. More recent models such as GeoLLaVA-8K (Wang et al., 2025a), GeoGround (Zhou et al., 2024b), and GeoPixel (Shabbir et al., 2025) have pushed the field toward ultra-high-resolution processing, unified visual grounding, and pixel-level grounding, showing a clear move from coarse scene description to finer spatial reasoning in remote sensing.

Despite the rapid progress of RS-MLLMs, most existing studies still emphasize capability expansion and benchmark construction rather than reliability analysis. Current evaluations (Hu et al., 2025b; Bazi et al., 2024; Zhan et al., 2025; Soni et al., 2025; Zhou et al., 2024b) measure task performance in remote sensing interpretation, but they provide limited insight into whether model responses are faithfully grounded in visual evidence. Recent work has linked hallucination in RS-MLLMs to weak grounding and insufficient localization (Liu et al., 2026; Zhou et al., 2026). However, the specific case of small-object hallucination remains underexamined, especially when queried targets are compact, densely distributed, and visually close to surrounding structures.

2.3. Small Object Detection

Small-object detection has long been a core topic in remote sensing image analysis, as reflected by widely used benchmarks such as DOTA (Xia et al., 2018) and DIOR (Li et al., 2020b). Early representative studies showed that progress on this problem depends heavily on stronger multi-scale representation and better handling of cluttered or clustered targets, as illustrated by SCRDet (Yang et al., 2019b), ClusDet (Yang et al., 2019a), and DMNet (Li et al., 2020a). Later work such as SCRDet++ (Yang et al., 2022) further improved robustness in cluttered scenes.

More surveys indicate that small-object detection is still difficult because fine-scale targets are strongly affected by scale imbalance, dense layouts, and background interference (Leng et al., 2024a; Hua and Chen, 2025). This conclusion is also supported by recent empirical analysis, which shows that even strong contemporary detectors can remain unstable on small objects from satellite imagery, especially under generalization settings (Yuan et al., 2026). At the method level, current research continues to refine feature preservation and context-aware representation for fine-scale targets, as seen in LARS (Li et al., 2024) and DCEDet (Hu et al., 2025a).

Existing small-object detection studies still center on specialized visual models and their ability to extract small targets from images. They do not explain how RS-MLLMs behave when small objects are queried through natural language. Whether these models can produce faithful small-object reasoning through natural-language interaction is still largely unexamined.

3. Benchmark

This section describes the construction of RS-SOH and its task-specific annotation design. The benchmark consists of 25,000 question–answer pairs derived from 1,000 image patches, covering 28 retained small-object categories. Each question is generated from object-level annotations and assigned to one of three diagnostic branches: existence reasoning, counting reasoning, or positional reasoning.

3.1. Task Definition and Hallucination Taxonomy

To evaluate small-object hallucination, RS-SOH organizes questions into three categories: existence, counting, and positional hallucination. These categories correspond to three increasingly demanding forms of visual grounding: deciding whether a queried object is present, verifying or producing its quantity, and supporting a spatial claim with object-level evidence.

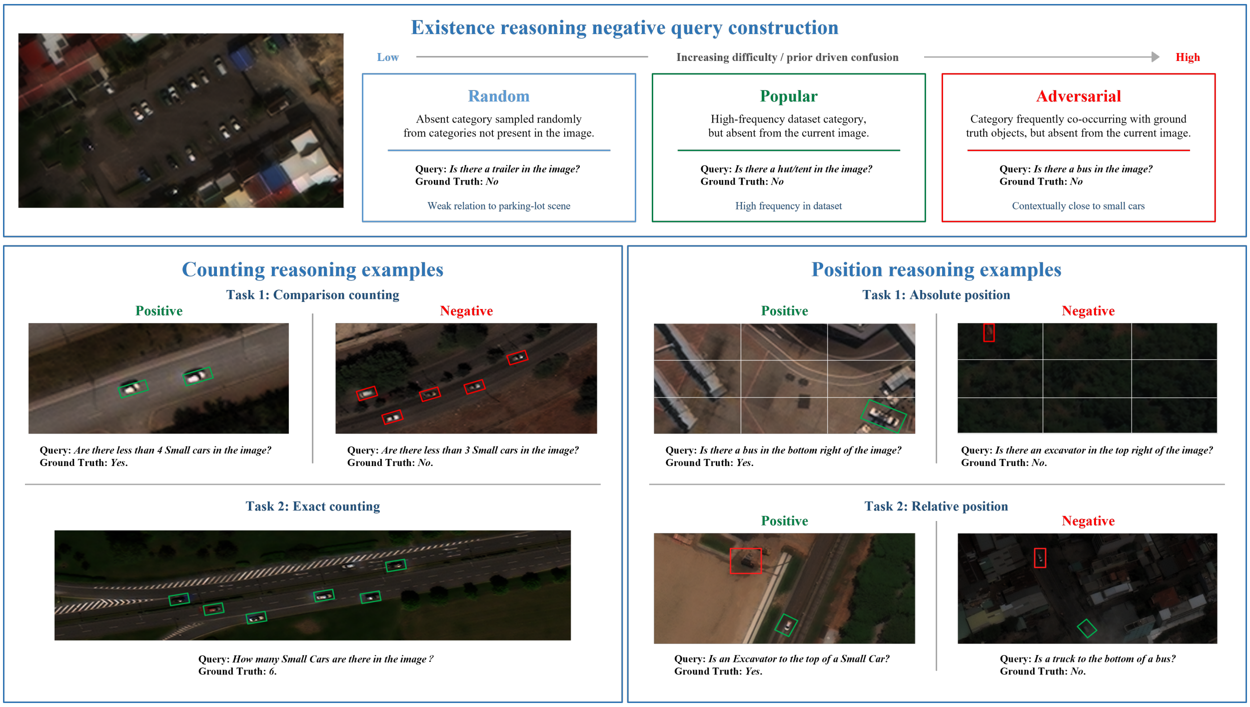

Fig. 1. Task examples in RS-SOH. The upper panel illustrates existence reasoning using presence category questions and three absent-category settings. The lower panels illustrate counting reasoning and positional reasoning, including their corresponding subtask forms.

Existence hallucination refers to an unsupported object-presence judgment. Given an image patch and a queried category, the model should answer according to object-level evidence. The existence branch contains object-presence questions and object-absence questions. Object-presence questions are derived from annotated categories, while object-absence questions are constructed under three settings, as shown in Fig. 1. *Random* samples a missing category weakly related to the current scene. *Popular* uses a frequent dataset category that is still absent from the image. *Adversarial* selects an absent category that is visually or contextually close to the annotated objects. These settings gradually increase the pressure of category frequency and scene plausibility on visual evidence.

Counting hallucination refers to an unsupported quantity judgment. The counting branch includes comparison counting and exact counting. Comparison counting asks the model to verify a numerical statement about the target category, whereas exact counting requires the model to output the number of target instances directly. Compared with existence reasoning, counting places stronger demand on instance separation because neighboring small objects, weak boundaries, and repeated background patterns can easily lead to missed or duplicated counts.

Positional hallucination occurs when a spatial claim is not supported by the image. The positional branch includes absolute-position reasoning and relative-position reasoning. Absolute-position reasoning links a target category to a predefined image region, while relative-position reasoning compares the target location with a reference object. This task tests whether the model can keep object identity connected to location rather than relying on a plausible scene-level layout.

At the task level, RS-SOH contains 25,000 question–answer pairs. These include 18,000 existence reasoning questions, 5,000 counting reasoning questions, and 2,000 positional reasoning questions. The positional branch is further divided into 1,246 absolute-position questions and 754 relative-position questions. These task-specific annotations are generated from the same retained image-patch and small-object annotation pool.

3.2. Dataset Collection and Benchmark-Specific Annotation

RS-SOH is constructed from xView, a large-scale overhead remote sensing dataset with high-resolution imagery and object-level annotation information (Lam et al., 2018). Since the original xView annotations are mainly designed for object detection, RS-SOH converts the imagery and instance-level annotations into patch-level records and task-specific question–answer labels for existence reasoning, counting reasoning, and positional reasoning.

Fig. 2 summarizes the construction pipeline of RS-SOH. Large source images are first cropped into patch-level samples, and object annotations are then matched to the corresponding patches. We further remove patches with insufficient category diversity and screen the remaining objects according to a predefined small-object scale threshold. Category filtering is then applied, and the retained small-object records are used to generate task-specific questions for existence, counting, and positional reasoning.

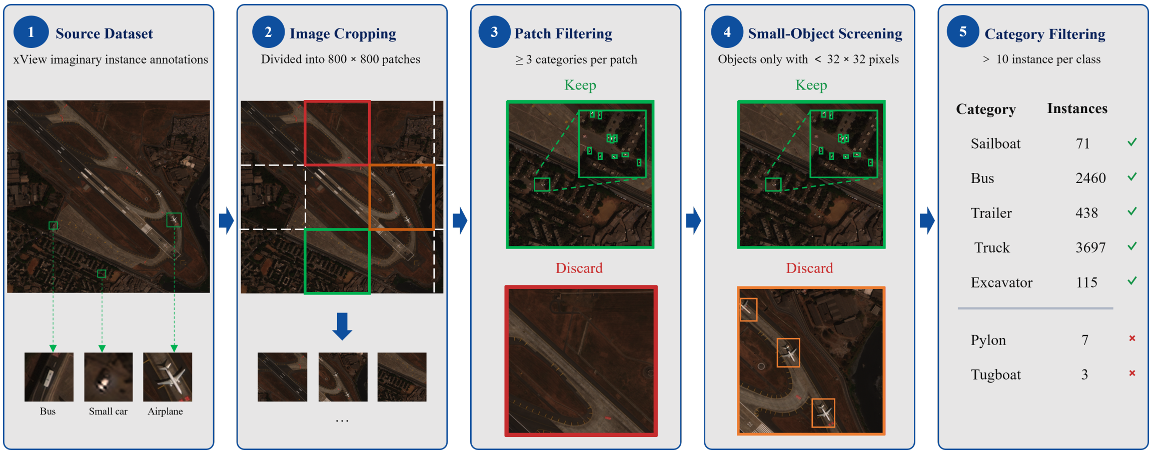

Fig. 2. Construction workflow of RS-SOH from xView imagery to the retained small-object pool. Green annotations denote retained patches or valid small-object instances that satisfy the filtering criteria, while red and orange annotations denote discarded patches removed during patch-level filtering.

The original source images are very large and usually contain multiple scene regions and many object instances. Such images are not suitable as direct prompt-based evaluation units because excessive visual content may introduce uncontrolled ambiguity and make question construction difficult to standardize. We first crop the images into 800×800 image patches. During cropping, the object-level annotations are spatially matched to the corresponding image patches, and invalid or unsuitable bounding boxes are removed. This process converts the original detection-style annotations into patch-level object records, which provide the basis for subsequent question–answer generation.

Following patch extraction, we apply two filtering steps. The first step is patch filtering. Patches containing fewer than three object categories are removed to avoid visually trivial samples and to ensure that each retained patch supports diverse positive, negative, counting, and spatial-relation questions. The second step is object-level screening. Since RS-SOH is designed to evaluate small-object hallucination, we retain only object instances whose bounding-box area is smaller than 32×32 pixels, following the COCO small-object definition (**?**). This criterion ensures that the benchmark focuses on compact targets rather than general object recognition. Category filtering is then used to remove categories with too few remaining small-object instances, improving the stability of category-level evaluation. The resulting benchmark image pool contains 1,000 retained image patches and 28 small-object categories.

Based on the filtered image-patch and object-record pool, we generate benchmark-specific labels for each task. For existence reasoning, category-presence records are used to define positive labels, while absent categories are sampled to form negative queries under random, popular, and adversarial settings. For counting reasoning, category-wise bounding-box counts are used to produce both binary comparison labels and exact-count labels. For positional reasoning, bounding-box centers are mapped to predefined image regions and pairwise spatial relations, producing labels for absolute-region and relative-relation questions.

3.3. Data Statitics

To illustrate the sample support of the retained categories, we count the number of small-object instances in each category and sort the categories in descending order. As shown in Fig. 3, the retained categories show a clear long-tailed distribution. Small Car has the largest number of instances, followed by vehicle-related categories such as Truck, Passenger Vehicle, Bus, Cargo Truck, and Utility Truck. Several categories have relatively fewer instances, such as Truck w/Liquid, Scraper/Tractor, and Ground Grader.

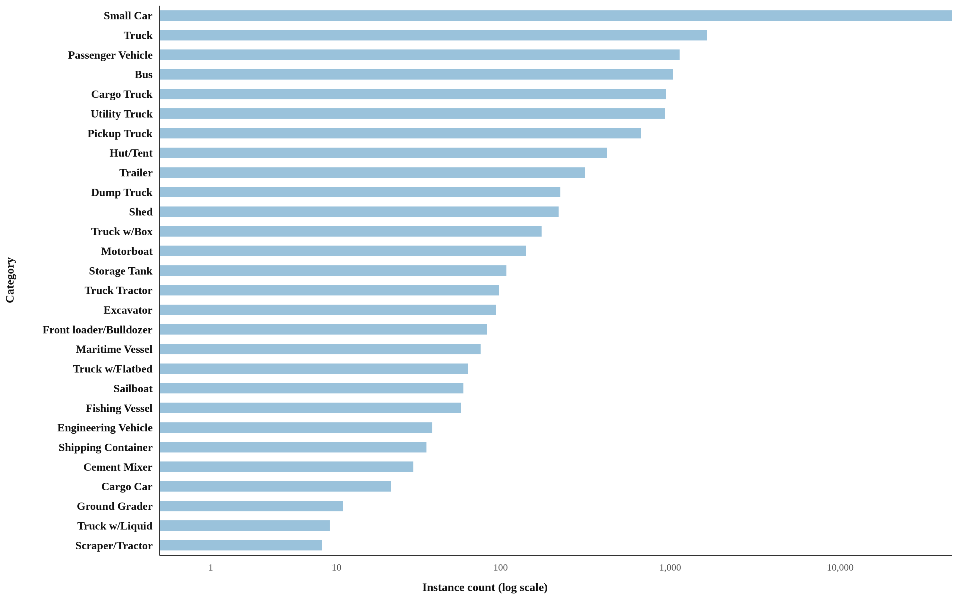

Fig. 3. Distribution of retained small-object instances across categories. Categories are sorted in descending order by instance count, and the x-axis is shown on a logarithmic scale. The distribution reflects the sample support of different categories in RS-SOH and reveals a long-tailed category structure.

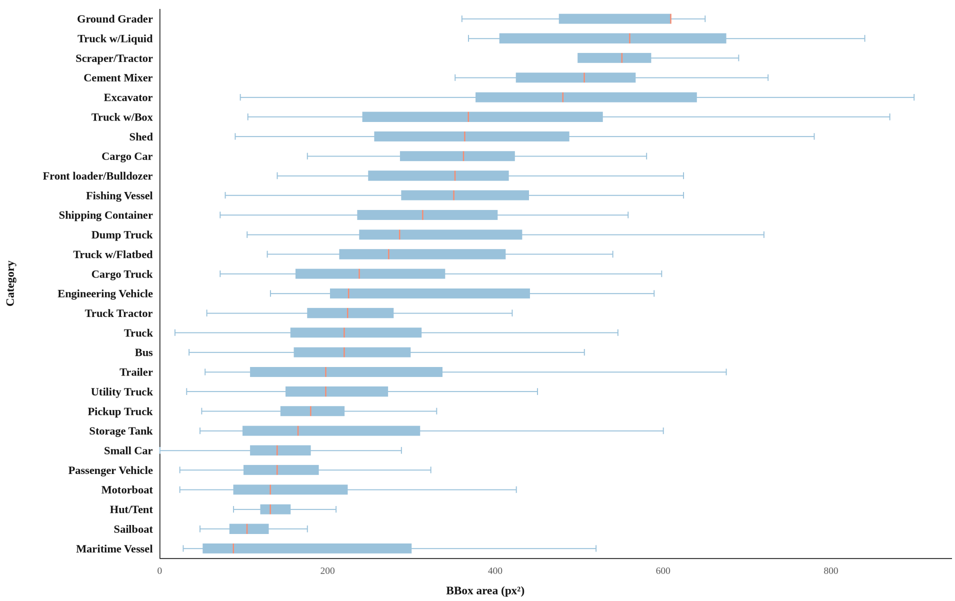

Fig. 4. Object-scale distribution of retained small-object categories. Categories are sorted in descending order by mean bounding-box area, and error bars denote the standard deviation of object area within each category. Although all retained objects satisfy the small-object criterion, clear scale differences remain across categories.

In addition to instance counts, we further analyze the object-scale distribution of the retained categories. As shown in Fig. 4, all retained categories fall within the small-object regime, but their mean bounding-box areas still vary substantially. For example, Truck w/Liquid, Ground Grader, Scraper/Tractor, and Cement Mixer have relatively larger mean object areas, while Small Car, Sailboat, Passenger Vehicle, and Maritime Vessel have relatively smaller mean object areas. The error bars represent the standard deviation of object area within each category, reflecting intra-category scale variation. This category-level scale information is further analyzed in Section 5 to examine how object size affects model performance across different reasoning tasks.

3.4. Evaluation Protocol and Metrics

We adopt a unified evaluation protocol across all models and question types. Each test sample contains one image patch and one natural language query. Model outputs are normalized according to the required answer format: binary questions are parsed as yes or no, and exact count questions are parsed as integers.

Binary questions include existence reasoning, comparison counting, and positional reasoning. We report Precision, Recall, and F1 score for these questions. Let TP, FP, and FN denote true positives, false positives, and false negatives, respectively. Precision measures the proportion of predicted positive answers that are correct. Recall measures the proportion of true positive cases that are successfully identified. F1 score is the harmonic mean of Precision and Recall and is used as the primary comparison metric for binary questions:

$$Precision = \frac{TP}{TP + FP}, \tag{1}$$

$$Recall = \frac{TP}{TP + FN}, \tag{2}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}. \tag{3}$$

These metrics capture different response patterns. High Recall with low Precision indicates that the model accepts many positive claims, including unsupported ones. High Precision with low Recall reflects a conservative response pattern, in which truly existing objects or spatial relations may be missed. Existence reasoning results are organized under the Random, Popular, and Adversarial settings. Within each setting, the binary metrics are computed over both present category questions and absent category questions, so the results capture missed objects as well as unsupported object presence claims. Comparison counting and positional reasoning use the same metrics. Exact counting is evaluated with an exact match criterion: a prediction is correct only when the generated integer matches the ground truth number of target instances. In positional reasoning, the absolute region and relative relation subsets are evaluated separately to distinguish object to image grounding from object to object spatial grounding.

4. Benchmarking RS-MLLMs

4.1. Evaluated Models

We evaluate ten representative RS-MLLMs: RS-LLaVA (Bazi et al., 2024), GeoChat (Kuckreja et al., 2024), LHRS (Muhtar et al., 2024), SkyEyeGPT (Zhan et al., 2025), TEOChat (Irvin et al., 2025), EarthDial (Soni et al., 2025), EarthMind (Shu et al., 2025), GeoPix (?), Falcon (Yao et al., 2025), and VHM (Pang et al., 2025). These models use different remote-sensing adaptation strategies, so their responses provide a useful test bed for examining how small-object errors change across different tasks.

4.2. Task-specific Performance on Small-Object Hallucination

4.2.1. Existence Reasoning

Table 1 reports the existence reasoning results under the three negative-query settings. Accuracy separates calibrated discrimination from recall-dominated answering. GeoChat, EarthDial, TEOChat, EarthMind, and GeoPix keep recall close to 100% while precision stays near 50%, so their F1 scores mainly come from accepting most queried categories as present. Accuracy remains close to chance because the same permissive pattern recovers present categories and also accepts many absent ones.

Table 1
Existence reasoning results on RS-SOH under Random, Popular, and Adversarial negative-query settings.

Strategy	Model	Accuracy	Precision	Recall	F1 Score
Random	GeoPix	51.03	50.53	99.00	66.91
	Falcon	61.40	62.45	57.17	59.69
	VHM	64.75	59.20	94.90	**72.92**
	EarthMind	52.15	51.10	99.93	67.62
	RS-LLaVA	67.53	**77.00**	50.00	60.63
	SkyEyeGPT	**68.38**	71.53	61.07	65.89
	LHRS	62.52	62.50	62.57	62.54
	TEOChat	50.48	50.24	99.93	66.87
	GeoChat	50.00	50.00	**100.00**	66.67
	EarthDial	50.22	50.11	99.97	66.76
Popular	GeoPix	49.83	49.92	99.00	66.37
	Falcon	52.33	52.13	57.00	54.46
	VHM	48.20	49.07	94.90	64.69
	EarthMind	49.97	49.98	99.93	66.64
	RS-LLaVA	**65.00**	**71.84**	49.33	58.50
	SkyEyeGPT	58.83	58.45	61.10	59.75
	LHRS	51.42	51.14	63.40	56.62
	TEOChat	49.97	49.98	99.93	66.64
	GeoChat	50.00	50.00	**100.00**	**66.67**
	EarthDial	49.98	49.99	99.97	66.65
Adversarial	GeoPix	49.85	49.92	99.00	66.38
	Falcon	52.33	52.13	57.07	54.49
	VHM	48.75	49.35	94.90	64.93
	EarthMind	50.02	50.01	99.97	**66.67**
	RS-LLaVA	**61.98**	**66.99**	47.99	55.92
	SkyEyeGPT	58.18	57.73	61.13	59.38
	LHRS	51.67	51.36	63.17	56.65
	TEOChat	49.98	49.99	99.93	66.64
	GeoChat	50.00	50.00	**100.00**	**66.67**
	EarthDial	50.02	50.01	99.97	66.66

RS-LLaVA, SkyEyeGPT, Falcon, VHM, and LHRS apply stricter positive judgments. SkyEyeGPT reaches the highest accuracy in the *Random* subset, while RS-LLaVA reaches the highest accuracy in both the *Popular* and *Adversarial* subsets. This pattern indicates stronger rejection of unsupported category claims, but the lower recall shows that some truly present small-object categories are also filtered out.

Performance generally becomes less stable when absent categories are frequent or contextually plausible. This change suggests that several models still allow category prior or scene plausibility to affect object-presence judgment when small-object evidence is weak.

4.2.2. Counting Reasoning

Table 2 reports comparison counting, and Fig. 5 reports exact counting. In comparison counting, GeoPix and TEOChat obtain the two highest F1 scores, 67.70% and 67.10%, mainly because they recover most positive numerical statements. Their accuracies are lower than EarthMind and EarthDial, indicating that high recall does not necessarily reflect stronger numerical discrimination.

Table 2
Comparison counting results on RS-SOH.

Model	Acc.	Prec.	Rec.	F1
RS-LLaVA	51.10	53.03	19.25	28.25
GeoChat	54.20	53.14	71.10	60.82
Falcon	51.78	53.00	31.70	39.66
VHM	57.40	58.43	51.30	54.63
TEOChat	55.58	53.28	90.60	67.10
EarthMind	**59.80**	62.10	50.30	55.58
LHRS	54.10	**69.07**	14.85	24.44
GeoPix	53.78	52.03	**96.90**	**67.70**
EarthDial	57.58	56.41	66.70	61.12
SkyEyeGPT	51.95	55.00	21.45	30.86

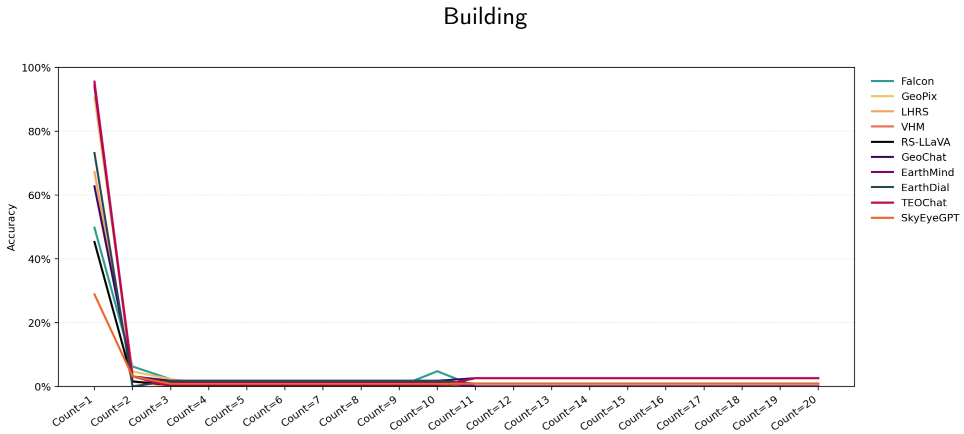

Fig. 5. Exact-match counting rate across different ground-truth count ranges on RS-SOH.

EarthMind has the highest accuracy, 59.80%, with 62.10% precision and 50.30% recall. This combination reflects a more balanced verification pattern: the model rejects more unsupported numerical claims while retaining a moderate fraction of true positives. VHM and EarthDial show similar but weaker balance. LHRS reaches the highest precision, 69.07%, but its recall is only 14.85%, showing a strongly conservative response pattern. RS-LLaVA, SkyEyeGPT, and Falcon show similar conservatism, which protects precision but weakens F1.

Exact counting exposes a sharper limitation. Fig. 5 shows that exact matching is mainly successful when the ground-truth count is one. The sharp drop at Count=2 indicates that many outputs still behave like single-object recognition, and the near-collapse in higher count ranges reflects difficulty in separating dense small-object instances.

4.2.3. Positional Reasoning

Tables 3 and 4 separate positional reasoning into absolute-position and relative-position questions. RS-LLaVA is excluded because its positional outputs are frequently non-compliant.

Table 3
Absolute-position reasoning results on RS-SOH.

Model	**Acc.**	**Prec.**	**Rec.**	**F1**
EarthMind	52.65	51.38	98.72	**67.58**
GeoPix	51.61	50.82	99.52	67.28
EarthDial	50.64	50.32	99.68	66.88
TEOChat	51.85	50.97	97.11	66.85
GeoChat	50.00	50.00	**99.84**	66.63
VHM	54.98	53.02	87.32	65.98
SkyEyeGPT	49.76	49.88	96.31	65.72
Falcon	**55.06**	**57.24**	39.97	47.07
LHRS	49.92	49.72	14.13	22.00

Table 4
Relative-position reasoning results on RS-SOH.

Model	**Acc.**	**Prec.**	**Rec.**	**F1**
EarthDial	50.00	50.00	**92.57**	**64.93**
GeoPix	49.60	49.73	74.54	59.66
GeoChat	**51.46**	51.14	65.25	57.34
EarthMind	**51.46**	51.34	55.70	53.44
VHM	51.19	51.81	34.22	41.21
Falcon	50.53	51.33	20.42	29.22
TEOChat	50.00	50.00	13.00	20.63
LHRS	50.53	**52.27**	12.20	19.78
SkyEyeGPT	50.40	52.00	10.34	17.26

In absolute-position reasoning, Falcon reaches the highest accuracy and precision, but its recall drops to 39.97%. Falcon therefore checks regional evidence more strictly, but misses many true region-object matches. EarthMind, GeoPix, EarthDial, TEOChat, and GeoChat obtain higher F1 scores through high recall, while their accuracies remain close to 50–53%. This means that broad region prompts still allow many unsupported spatial confirmations. VHM has the second-highest accuracy, 54.98%, and keeps higher recall than Falcon, indicating a less restrictive regional checking pattern.

Relative-position reasoning has a narrow accuracy range, from 49.60% to 51.46%, despite large differences in recall and F1. EarthDial achieves the highest F1 score by recovering many positive relations, but its 50.00% accuracy and 50.00% precision indicate weak separation between true and false relations. GeoChat and EarthMind obtain the highest relative-position accuracy, 51.46%, but the margin is small, so the task remains close to chance-level discrimination.

The contrast between the two subsets shows that pairwise relation reasoning is harder than region-level grounding. Relative-position questions require the model to localize two object categories and preserve relation direction, so an error in either object localization or relation comparison can flip the final answer.

4.3. Cross-Task Model Disparity Analysis

To investigate model-level disparities across reasoning tasks, we use the cross-model distribution of F1 scores as the comparison basis. Fig. 6 shows a compact distribution for existence reasoning, where most models fall within the low-to-mid 60% range. Such compression suggests that category-level recognition and coarse scene matching provide limited separation of model-level grounding ability.

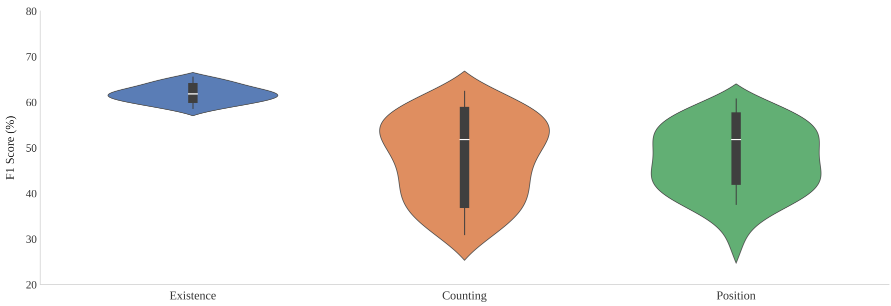

Fig. 6. Distribution of F1 scores across RS-MLLMs under existence reasoning, counting reasoning, and positional reasoning.

Counting reasoning spreads the models more strongly, with F1 scores ranging from the 20% range to above 60%. The wider distribution suggests that numerical verification depends on object-wise evidence rather than only category plausibility. Models that better retain instance-level cues remain near the upper end, whereas models relying on coarse semantic impressions drop to the lower end.

Positional reasoning also produces a wide gap across models, but the separating factor shifts from enumeration to spatial binding. The model must attach the queried category to the correct region or reference object before verifying the spatial relation. These results show that counting and positional reasoning are more diagnostic than existence reasoning because they require object-level evidence beyond category presence.

5. Scale-Aware Performance Analysis

This section examines whether object scale affects small object hallucination in RS-MLLMs. Object scale is measured at the category level using the mean bounding box area of retained small object instances. For each reasoning task, prediction results are aggregated by object category, and category level accuracy is paired with the corresponding object scale. Spearman correlation is then used to test whether larger object categories tend to produce better model performance.

5.1. Scale vs Performance

Table 5 summarizes the correlation between object scale and task accuracy. Existence reasoning shows only a weak and non-significant correlation with object scale, and positional reasoning shows a similar pattern. Counting is the only task with a significant positive correlation, suggesting that larger objects within the small-object regime may be easier for RS-MLLMs to count correctly. This task difference may be related to the visual operation required by each reasoning type. Existence reasoning can be influenced by category-level cues and scene context, so moderate scale variation may not strongly change a binary presence judgment. Positional reasoning depends on whether the object can be linked to a region or reference object, and this spatial binding step may still fail even when the target is slightly larger. Counting may be more sensitive to object scale because the model needs to separate neighboring instances before producing or verifying a number. When targets are compact, densely arranged, or weakly separated from the background, the model is more likely to merge adjacent objects, miss subtle instances, or produce unstable counts.

Table 5
Overall Spearman correlation between object scale and task accuracy.

Task	Spearman ρ
Existence	0.1363
Counting	0.3871*
Position	0.0930

* indicates statistical significance at $p < 0.05$.

5.2. Model-Dependent Scale Effects in Counting

Table 6 reports the model-level Spearman correlation between object scale and counting accuracy. EarthDial, EarthMind, and Falcon show significant positive correlations, suggesting that their counting predictions may remain partly related to object size. For these models, objects with larger sizes may provide clearer boundaries and more separable visual evidence, which could help numerical verification. Other models show weak or non-significant correlations, indicating that the overall scale–accuracy relationship in counting is not shared equally by all RS-MLLMs. A weak model-level correlation should not be directly interpreted as robustness to object scale, since counting errors may also be influenced by dense layouts, category confusion, missed instances, over-counting, or response calibration. These model-level differences suggest that object scale is more likely to affect counting when the model can use visual evidence to separate target instances; otherwise, counting outputs may become less consistently related to the physical size of the target.

Table 6
Model-level Spearman correlation between object scale and counting accuracy.

Model	Spearman ρ
EarthDial	0.3795*
EarthMind	0.4344*
Falcon	0.3678*
GeoChat	0.0657
GeoPix	-0.1161
LHRS	0.1115
RS-LLaVA	0.1886
TEOChat	-0.1150
VHM	0.0640
SkyEyeGPT	-0.1648

* indicates statistical significance at $p < 0.05$.

6. Discussion

RS-SOH evaluates small-object hallucination in RS-MLLMs by linking each question to object-level evidence. The benchmark uses remote-sensing image patches and annotations that specify object presence, object count, and spatial relation labels. This setting allows unsupported responses to be examined beyond category naming alone. The existence-reasoning results show that several models recover many positive cases but also accept some absent categories as present, as reflected by high recall and much lower precision. Counting introduces a different difficulty: model performance decreases when the answer requires numerical verification or exact enumeration, especially when several small objects appear in one patch. Positional reasoning further shows that recognizing a category does not ensure a supported spatial answer, since the queried object still needs to be linked to the correct region or reference object.

The existence, counting, and positional tests all point to a shared difficulty: small objects provide limited visual evidence for reliable verification. Small targets occupy limited image areas and can resemble nearby structures, which makes object checking difficult. When the queried object is not clearly grounded, the model may rely more on category plausibility or scene context. This interpretation is supported by the high-recall and low-precision behavior in existence reasoning, the unstable numerical outputs in counting reasoning, and the weak spatial grounding observed in positional reasoning. The scale-aware analysis provides additional evidence. Counting is the only branch with a significant positive correlation between object scale and task accuracy, suggesting that retained objects with larger bounding-box areas may provide more usable visual evidence for separating neighboring instances.

RS-SOH can be further extended to cover a broader range of small-object hallucination settings. Future versions could include additional target categories, more fine-grained reasoning forms, or remote-sensing data from other sensors or imaging conditions. Such extensions would help examine whether the hallucination behaviors observed in RS-SOH remain stable when object types, spatial relations, and visual characteristics become more diverse.

The RS-SOH results point to a mitigation direction that combines data-level design with region-aware model inference. In existence reasoning, high recall together with lower precision means that models recover many present categories but also accept plausible absent categories. Mitigation at the instruction-tuning stage can therefore include harder absent-category questions whose queried targets fit the scene context but are not present in the image, following negative-instruction and normal–misleading designs (Liu et al., 2024; Zhou et al., 2026). Related hallucination studies provide a similar explanation: hallucination can be reduced when decoding or revision gives greater weight to visual evidence rather than language priors (Leng et al., 2024b; Zhou et al., 2024a). This suggests that harder absent-category questions in RS-SOH should work as evidence-verification cases, rather than as negative samples alone. In counting comparison and exact count settings, unstable numerical answers are closely tied to nearby small targets that are not clearly separated. In absolute and relative position settings, unsupported spatial answers occur when the queried object or reference object is not anchored to a verified image region. Recent remote-sensing hallucination mitigation work provides a useful two-step evidence process for this type of problem (Liu et al., 2026). The first step is where-oriented evidence acquisition, in which the model localizes candidate regions related to the queried category before answering. The second step is what-oriented evidence refinement, in which the model checks local details within the selected region to distinguish nearby small targets and verify the relevant object evidence. MARINE (Zhao et al., 2024) provides a useful implementation reference for this step: before generating the final answer, the model can use image-grounded guidance to extract object-level information from the candidate region. This checking logic can also be connected to Woodpecker (Yin et al., 2024) more specifically: its key concept extraction and question formulation steps provide a useful way for remote-sensing models to decompose a visual query into verifiable elements, such as the queried category, candidate object region, object count, and reference object. For remote-sensing models, this means that a counting or positional answer should not be generated directly from the whole image. Instead, the model should first verify the local object evidence and then use the verified region or object pair to constrain the final response. The checked region or object pair can then be passed to the language decoder, so that counting and positional answers are generated from verified small-object evidence rather than scene-level plausibility alone. Future RS-MLLMs may therefore need to keep regional evidence involved throughout reasoning, allowing the final response to be constrained by checked object evidence before it is expressed in language.

7. Conclusion

RS-MLLMs have shown strong potential for remote-sensing image understanding, but their reliability in small-object scenarios remains insufficiently evaluated. This paper introduces RS-SOH, a benchmark for evaluating small-object hallucination in RS-MLLMs. The experiments show that weak image evidence allows scene-level associations to shape the answer before the queried object is verified. Small targets increase this risk because their boundaries and individual instances are difficult to separate. The scale-aware analysis indicates that object size affects model behavior most clearly when the answer depends on instance separation. Future work can use RS-SOH to test whether existing hallucination mitigation methods reduce errors on small objects. By comparing the same RS-MLLMs before and after mitigation, the evaluation can examine whether improvements occur in object-level localization, instance separation, and spatial verification, rather than only in whole-image response quality. This would show whether mitigation methods can address the weak-evidence failure chain observed in small-object hallucination.

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